EDF with constrained deadlines (Baruah et al, 1990)

Processor demand criterion

A set of n periodic tasks $\{\tau_1, \ldots, \tau_n\}$ with $D_i \leq T_i$ $\forall$ task τ_i is schedulable by EDF if and only if in any interval of time $[t_1, t_2]$ the processor demand $g(t_1, t_2)$ does not exceed the available time, i.e., $g(t_1, t_2) \leq t_2 - t_1$ $\forall\, t_1, t_2$ with $t_1 < t_2$ 8.

IN TEORIA SONO INFINITI. DOBBIAMO STUDIARE

- The processor demand in the time interval $[t_1, t_2]$ is the processing time requested by jobs activated in $[t_1, t_2]$ with absolute deadline $\leq t_2$: CIOÈ CHE HANNO GARANZIA DI TERMINARE ENTRO t_2.

$$g(t_1, t_2) = \sum_{i=1}^{n} \eta_i(t_1, t_2) C_i$$

where $\eta_i(t_1, t_2)$ is the no. of jobs of τ_i contributing to demand in $[t_1, t_2]$:

- $\eta_i(t_1, t_2) := |\{\tau_{i,k} \mid a_{i,k} \in [t_1, t_2] \wedge d_{i,k} \leq t_2\}| = \max\{0, K_2^i - K_1^i\}$ NON SI CONSIDERANO I JOB LA CUI DEADLINE È DOPO t_2.
- $K_2^i := |\{\tau_{i,k} \mid a_{i,k} \in [\phi_i, t_2] \wedge d_{i,k} \leq t_2\}| = \lfloor (t_2 + T_i - D_i - \phi_i)/T_i \rfloor$ FASE, MOMENTO DI ARRIVO DEL PRIMO JOB. TUTTI I JOB CHE MI INTERESSANO PIÙ QUELLI RILASCIATI PRIMA DI t_2.
- $K_1^i := |\{\tau_{i,k} \mid a_{i,k} \in [\phi_i, t_1]\}| = \lceil (t_1 - \phi_i)/T_i \rceil$ I JOB RILASCIATI PRIMA DI t_2.

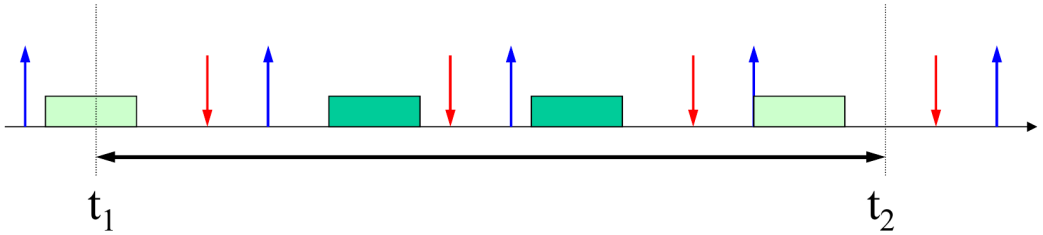


Image by Prof. G. Buttazzo, Scuola Superiore Sant'Anna, Pisa

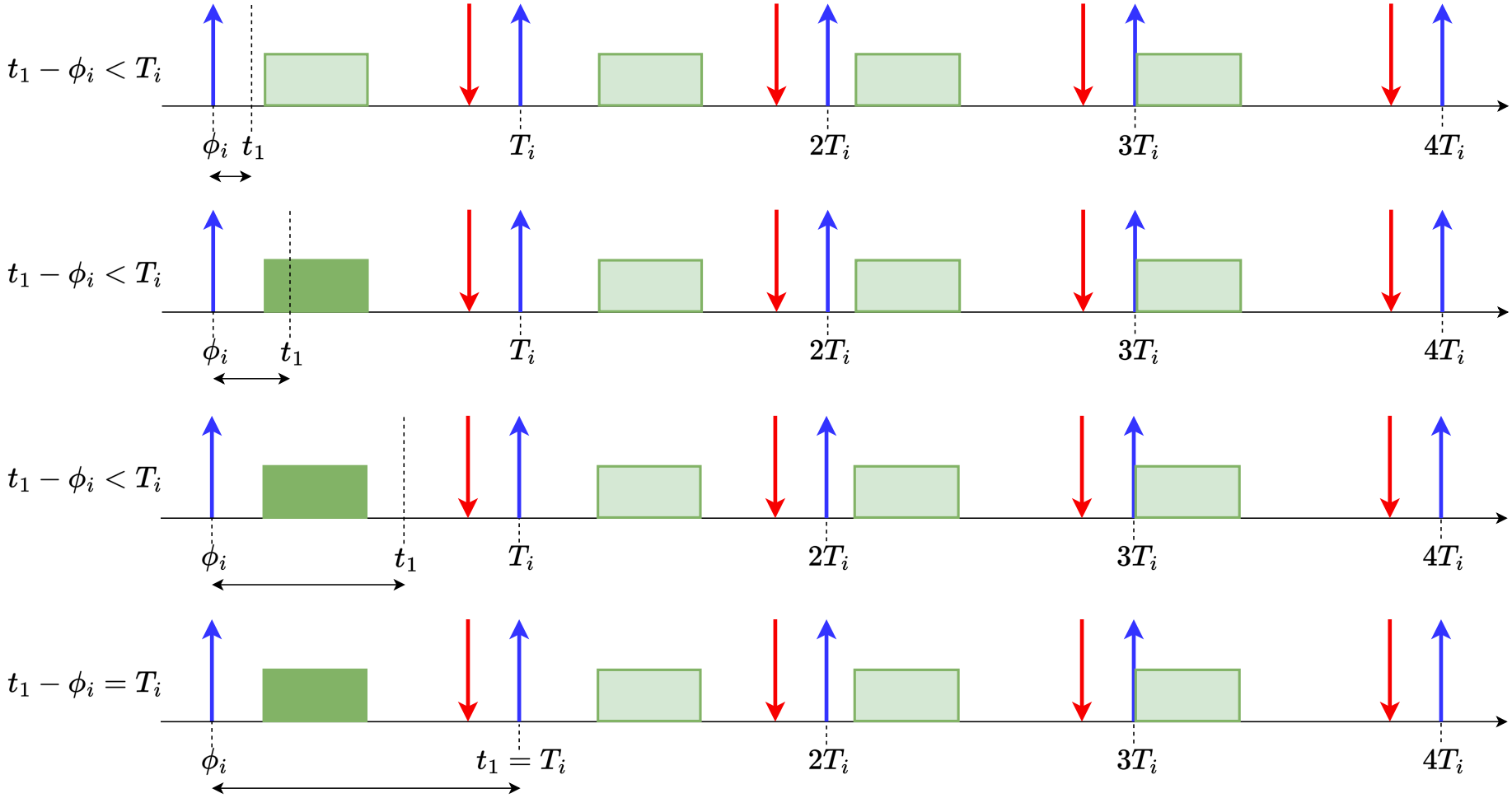
S. K. Baruah, L. E. Rosier, R. R. Howell, "Algorithms and complexity concerning the preemptive scheduling of periodic, real-time tasks on one processor", Journal of Real-Time Systems, 2, 1990

EDF with constrained deadlines: evaluation of K_1

CON $\lceil \cdot \rceil$ PRENDO ANCHE QUELLI CON DEADLINE DOPO t_1.

- $K_1^i := |\{\tau_{i,k} \mid a_{i,k} \in [\phi_i, t_1]\}| = \lceil (t_1 - \phi_i)/T_i \rceil$

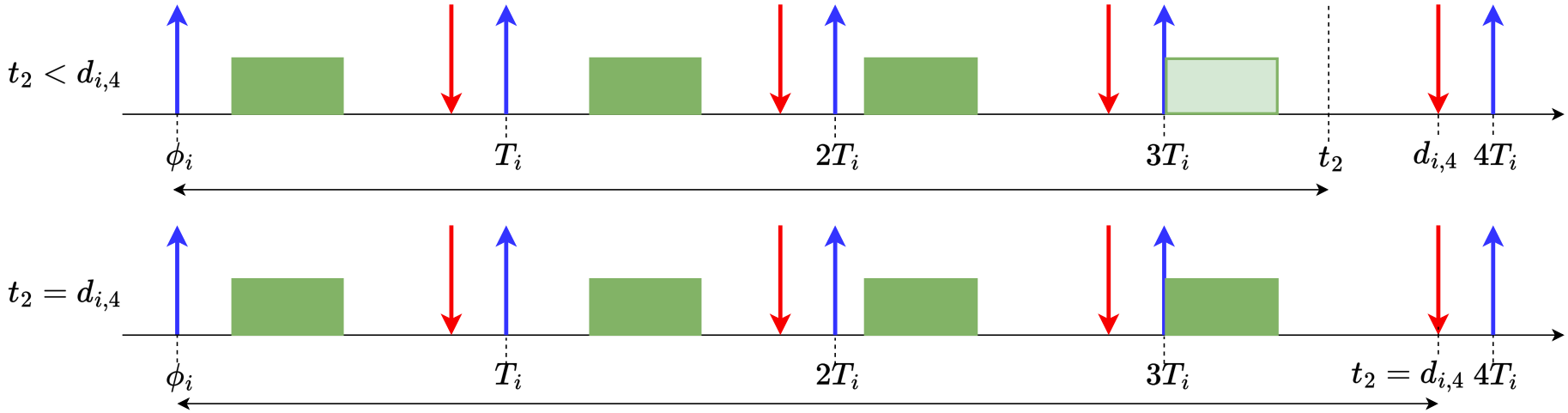
È IL # DI τ_i RILASCIATI NEL PERIODO $t_1 - \phi_i$



EDF with constrained deadlines: evaluation of K_2

SERVE PER AGGIUNGERE L'ULTIMO JOB NEL CASO $t_2 = d_{i,4}$
FA SI CHE AL NUM SI AGGIUNGA
UN PEZZO CHE LO FA DIVENTARE UN INTERO DEL DENUM

- $K_2^i := \left|\{\tau_{i,k} \mid a_{i,k} \in [\phi_i, t_2] \wedge d_{i,k} \leq t_2\}\right| = \lfloor t_2 + T_i - D_i - \phi_i / T_i \rfloor$



EDF with constrained deadlines: demand bound function

- Worst case scenario: all tasks activated at time 0 (i.e., $\phi_i = 0 \ \forall$ task τ_i):

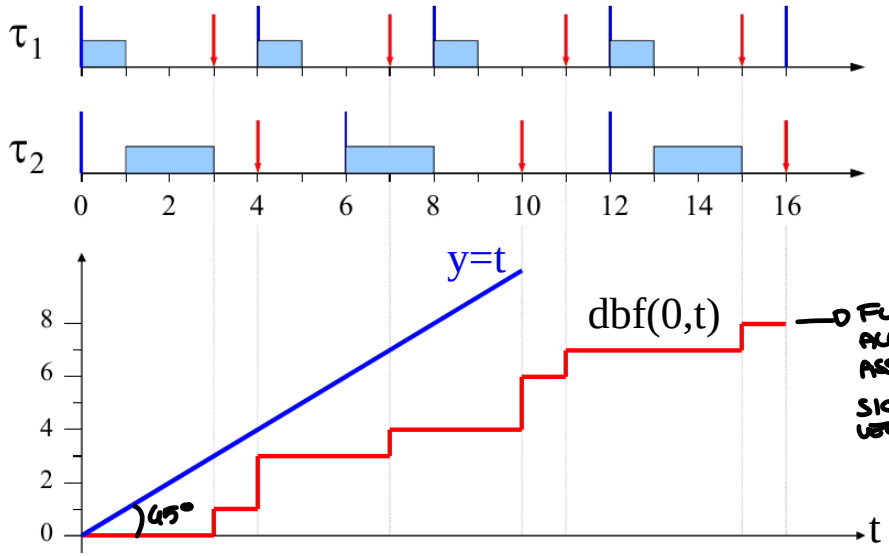
DEMAND BOUND FUNCTION

L'ISTANTE INIZIALE È 0

$$\text{dbf}(t) := g(0,t) = \sum_{i=1}^{n} \eta_i(0,t) C_i = \sum_{i=1}^{n} \left\lfloor \frac{t + T_i - D_i}{T_i} \right\rfloor C_i$$

Remark

A synchronous set of n periodic tasks $\{\tau_1, \ldots, \tau_n\}$ with $D_i \leq T_i \ \forall$ task τ_i is schedulable by EDF if and only if $\text{dbf}(t) \leq t \ \forall\, t > 0$



FUNZIONE A GRADINI CHE CRESCE ALLO SCOCCARE DI OGNI DEADLINE ASSOLUTA.

SICCOME È A GRADINO FARÒ LE MIE VERIFICHE ALLO SCOCCARE DELLE DEADLINE.

Image adapted from Prof. G. Buttazzo, Scuola Superiore Sant'Anna, Pisa